

Isothermal vapour/liquid equilibria of binary mixtures with dibutyl ether at 298.15 K†

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A head-space gas chromatographic technique has been used to determine vapour/liquid equilibria at 298.15 K of binary mixtures of di-n-butyl ether plus an organic compound (hexane, octane, diethyl ether, tetrahydrofuran, propylamine, butylamine, propanone, cyclopentanone, methanol, butan-1-ol, acetonitrile). Excess molar Gibbs energies, G^E , for the systems investigated have been obtained by a least-squares treatment of equilibrium data. Mixtures with hexane, octane and diethyl ether revealed an almost ideal behaviour. Other systems showed positive deviations, which increase with the polarity of the second component. Dibutyl ether + butanol and + cyclopentanone revealed the formation of azeotropic compositions. The Kirkwood–Buff integrals over the entire composition range have been calculated from the composition dependence of G^E , and briefly discussed.

Introduction

As a continuation of our studies on the thermodynamic properties of fluid mixtures^{1–4} with the aim of gaining insights into solute–solvent and solute–solute interactions, we herein report the results of vapour/liquid equilibrium (VLE) measurements for binary mixtures of di-n-butyl ether (DBE) (1) with each of the following organic compounds (2): hexane, octane, diethyl ether, tetrahydrofuran, propylamine, butylamine, propanone, cyclopentanone, methanol, butan-1-ol, acetonitrile. The second component was chosen so as to present a different polarity, size and shape.

Recently DBE has received particular attention because of the wide use of ethers as gasoline additives⁵ and of its specific use in the extractive distillation of azeotropic mixtures.⁶ Furthermore, DBE acquired some biological and pharmaceutical interest being included in a pool of solvents which yield partition coefficients with water correlated with the ability of the solutes to cross natural membranes (quantitative structure–activity relationships, QSAR).⁷ Though these correlations are usually established with the solvation properties, which refer to the infinite dilution limit, a thermodynamic investigation of binary systems including one of the solvents used in the QSAR procedures, that is DBE, over the whole composition range appears to be an interesting complementary approach. In addition, such an investigation on VLE of mixtures with DBE provides useful data for testing existing theories of solutions and for extending group contributions schemes, as well as to promote the general understanding of solute–solute–solvent interactions.

Isobaric VLE data for (DBE + diethyl ether)⁸ and (DBE + butan-1-ol),^{9–11} excess Gibbs energies at 298.15 K for (DBE + hexane),¹² as well as activity coefficients of DBE at infinite dilution in methanol (at 313.15 K) and in butanol (at

373.15 K)¹³ are reported in the literature. In addition we carried out isothermal VLE measurements for such mixtures in order to obtain an homogeneous set of results.

Excess Gibbs energies, G^E , and related quantities such as activity coefficients, γ_i , have been calculated by treating, with a method similar to that of Barker,¹⁴ VLE data collected by means of an equilibration cell coupled with a gas-chromatographic apparatus, described previously.^{15,16} From the composition dependence of γ_i the Kirkwood–Buff integrals¹⁷ in the whole composition domain have been computed and analysed to get information on the local structure of the examined mixtures with particular attention to the behaviour of the component at infinite dilution in DBE.

Experimental

The head-space gas chromatographic (GC) analysis for measuring the vapour-phase mole fraction, y_i , as a function of the liquid-phase mole fraction, x_i , was employed. The apparatus, composed of an equilibration cell and a gas chromatograph (C. Erba Mod. 5300 Mega), has already been described in detail elsewhere.^{15,16} An SE52 fused silica capillary column and a flame ionisation detector were used in all analyses.

All chemicals were commercial products of the best grade quality and were used without further purification. Their purity, in mass%, was checked by GC and their water content by Karl–Fischer titration. No significant difference was found with respect to the impurity content declared by the factory. Dibutyl ether, Aldrich 99.3%, water 0.005%; hexane, Fluka 99.5%; octane, Fluka 99.5%; diethyl ether, Riedel-de Haën 99.5%, water 0.1%; tetrahydrofuran, Fluka 99.5%, water 0.05%; propanone, Riedel-de Haën 99.5%, water 0.2%; cyclopentanone, Fluka 99%; propylamine, Fluka 99%, water 0.2%; butylamine, Aldrich 99.5%; methanol, C. Erba 99.9%, water 0.05%; butan-1-ol, C. Erba 99.5%, water 0.1%; acetonitrile, C. Erba 99.8%, water <0.1%. Thanks to the GC technique employed in determining the vapour composition, impurities

† Electronic Supplementary Information available. See <http://www.rsc.org/suppdata/cp/b0/b005160h/>

as large as those given above do not affect appreciably the reliability of the results.

All mixtures were prepared by weighing. The mole fractions, x_i , after correction for buoyancy and for evaporation of constituents in the vapour space of the container, were evaluated to better than 0.0003. At least three vapour samples were analysed for each given composition.

Data treatment

The values of vapour mole fractions of DBE, y_1 , were obtained by

$$y_1 = 1/[1 + r_f/(A_1/A_2)] \quad (1)$$

from the measured peak area ratio A_1/A_2 of the two components in the vapour phase and from the response factor r_f of the chromatograph for the mixture under consideration. This latter is given by

$$r_f = (A_1^*/A_2^*)/(p_1^*/p_2^*) \quad (2)$$

A_i^* being the peak area of the vapour equilibrated with the pure liquid i , having a vapour pressure p_i^* . The overall uncertainty in y_1 , resulting mostly from the uncertainty in A_1/A_2 , was generally less than 0.005 and, in the dilute regions, lower than 2% of y_1 of the less concentrated component.

The experimental A_1/A_2 values are related to the ratio of the experimental activity coefficients of the components, $(\gamma_1/\gamma_2)_{\text{exp}}$, according to

$$(\gamma_1/\gamma_2)_{\text{exp}} = (x_2/x_1)(A_1/A_2)(A_2^*/A_1^*)(W_1/W_2) \quad (3)$$

where W_i is the factor relevant to vapour phase non-ideality of component i , calculated accounting for the presence of air, assuming the latter as a third component equal to nitrogen. The expression for W_i is given elsewhere.¹⁸ Table 1 gives the values of the vapour pressures and molar volumes of the pure components, as well as the molar virial coefficients B_{ii} and B_{ij} used for the evaluation of W_i . Although some B_{ii} and most of B_{ij} are estimated values, probably affected by large uncertainties, the W_i terms usually differ from 1 by less than 4%.

As the function representing the ratio of calculated activity coefficients, $(\gamma_1/\gamma_2)_{\text{calc}}$, we found the Redlich–Kister equation convenient with at most three parameters

$$\ln(\gamma_1/\gamma_2)_{\text{calc}} = a_1(x_2 - x_1) + a_2(6x_1x_2 - 1) + a_3(x_1 - x_2)(8x_1x_2 - 1) \quad (4)$$

The coefficients a_k were obtained by a non-linear least-squares procedure, similar to that described by Barker¹⁴ for the treat-

ment of P - x data. In our case the objective function was

$$S = \sum_{k=1}^n [\ln(\gamma_1/\gamma_2)_{\text{exp}} - \ln(\gamma_1/\gamma_2)_{\text{calc}}]^2 \quad (5)$$

the sum S of the residuals being extended over all n experimental points. The ratio $A_2^*/A_1^* = p_2^*/p_1^*(1/r_f)$, which appears in eqn. (3), was treated as an adjustable parameter to be determined in the least-squares calculations. As a consequence, neither the direct determination of the response factor r_f via calibration experiments, nor the measurements of the vapour pressures p_i^* are necessary for obtaining activity coefficients.

Excess molar Gibbs energies G^E were calculated through the equation

$$G^E/RT = x_1x_2 \sum_{k=1}^3 a_k(x_1 - x_2)^{k-1} \quad (6)$$

Besides the actual experimental quantities y_i and γ_1/γ_2 , we can also obtain the following “mixed” quantity, γ_i^{exp} :

$$\gamma_i^{\text{exp}} = y_i^{\text{exp}} \sum_k (\gamma_k^{\text{calc}} x_k p_k^*/W_k)/(x_i p_i^*/W_i) \quad (7)$$

which contains the experimental datum y_i^{exp} and the calculated total pressure of the mixture (the summation). From the γ_i^{exp} a sort of experimental excess Gibbs energy G_{exp}^E and total pressure P_{exp} can be obtained:

$$G_{\text{exp}}^E = RT \sum_k x_k \ln \gamma_k^{\text{exp}} \quad (8)$$

$$P_{\text{exp}} = \sum_k \gamma_k^{\text{exp}} x_k p_k^*/W_k \quad (9)$$

Results and discussion

The experimental x_1 , $\ln(\gamma_1/\gamma_2)$, y_1 and P_{exp} quantities are given in Table S1, deposited as Electronic Supplementary Information (ESI).[†] The values of the a_k parameters of eqn. (4) are reported in Table 2 together with the values of the response factor r_f obtained from the fitting procedure (and from the literature p_i^* values given in Table 1) and the standard deviation σ for the representation of $\ln(\gamma_1/\gamma_2)$. The σ values did not exceed 0.04. The standard deviation on y was usually <0.005.

In Fig. 1 the plot of $\ln(\gamma_1/\gamma_2)$ vs. composition is shown for DBE + methanol and DBE + hexane. For the latter system, data from other authors are also reported for comparison.

Among the examined systems, DBE + cyclopentanone and DBE + butanol revealed the formation of azeotropic mixtures at 298.15 K, as illustrated in Fig. 2. The azeotropic composi-

Table 1 Vapour pressures p^* , molar volumes V^* , second virial coefficient B_{ii} for pure compounds and cross virial coefficients B_{ij} at 298.15 K

Compound	$p^*{}^a$ /kPa	$V^*{}^b$ /cm ³ mol ⁻¹	$-B_{ii}$ /dm ³ mol ⁻¹	$-B_{ij}$ ^c /dm ³ mol ⁻¹
Dibutyl ether	0.824 ^d	170.4	5.6 ^e	—
Hexane	20.15	131.5	1.95 ^f	2.6
Octane	1.86	163.5	3.81 ^g	3.4
Diethyl ether	71.56	104.7	1.24 ^g	2.0
Tetrahydrofuran	21.62	81.7	1.2 ^f	1.8
Propylamine	42.13	83.1	1.4 ^h	1.7
Butylamine	12.23 ^d	99.8	2.0 ^h	2.1
Propanone	30.81	74.0	2.0 ^f	1.5
Cyclopentanone	1.52	89.1	2.7 ⁱ	2.0
Methanol	16.96	40.7	1.94 ^g	0.8
Butan-1-ol	0.872 ^j	92.0	3.38 ^g	2.0
Acetonitrile	11.85	52.9	6.3 ^f	1.1
Nitrogen	—	—	0.0 ^f	0.1 ^k

^a Taken from Reid *et al.* (ref. 19). ^b Our data taken from ref. 20. ^c Computed according to Scatchard and Ticknor (ref. 21). ^d Taken from Gmehling *et al.* (ref. 22). ^e Estimated from C₂–C₆ ethers and C₁–C₈ alkanes. ^f From Dymond and Smith (ref. 23). ^g From Tsonopoulos and Dymond (ref. 24). ^h Estimated with the group contribution method by McCann and Danner (ref. 25). ⁱ Estimated from C₃–C₅ ketones and C₃–C₆ alkanes and cycloalkanes. ^j Taken from Gmehling *et al.* (ref. 26). ^k Ref. 23. The same value applies for all mixtures of nitrogen (or oxygen) with the listed substances.

Table 2 Chromatographic response factor r_f , parameters a_k of eqn. (4), and standard deviation σ for binary mixtures of dibutyl ether with various organic compounds at 298.15 K

Compound	r_f	a_1	a_2	a_3	σ^a
Hexane	4.64	0.0571	—	—	0.018
Octane	1.40	0.0678	—	—	0.008
Diethyl ether	5.90	0.0076	—	—	0.014
Tetrahydrofuran	4.17	0.1403	−0.0358	—	0.016
Propylamine	4.54	0.6067	0.1369	0.1620	0.039
Butylamine	2.25	0.5136	0.1730	0.1668	0.034
Propanone	8.19	1.0970	−0.1231	0.0874	0.015
Cyclopentanone	1.40	0.9620	−0.0400	0.1144	0.020
Methanol	9.62	2.0030	−0.0957	0.3719	0.026
Butan-1-ol	1.73	1.3777	0.2125	0.2043	0.023
Acetonitrile	4.97	2.0515	−0.2278	0.4049	0.019

^a Standard deviation of the fit on $\ln(\gamma_1/\gamma_2)$.

tion, evaluated by the equation

$$y_i = (\gamma_i x_i p_i^*/W_i) / \sum_k \gamma_k x_k p_k^*/W_k \quad (10)$$

using p_i^* values from Table 1, was $x_{\text{DBE}}(\text{az}) = 0.18$ and 0.52 for DBE + cyclopentanone and DBE + butanol systems, respectively.

The G^E curves computed by eqn. (6) are presented in Fig. 3 for the binary mixtures of DBE with tetrahydrofuran, propylamine, propanone, butanol, or acetonitrile. The curves are slightly asymmetric, with maxima at $0.4 < x_{\text{DBE}} < 0.6$. In Table 3 are given the values of the activity coefficients of the components at infinite dilution, γ_i^∞ , obtained from parameters of eqn. (4), as well as the excess Gibbs energies for equimolar mixtures, G^E ($x = 0.5$), and corresponding standard deviations. The table also reports data from other authors. The activity coefficients for DBE + hexane, obtained by Marsh *et al.*¹² from P - x - y measurements, agree with our results within our experimental error (see also Fig. 1). Literature data for DBE with diethyl ether, methanol, or butanol, being obtained from isobaric VLE^{8–11} and/or at temperatures far from room temperature,¹³ do not allow a proper comparison.

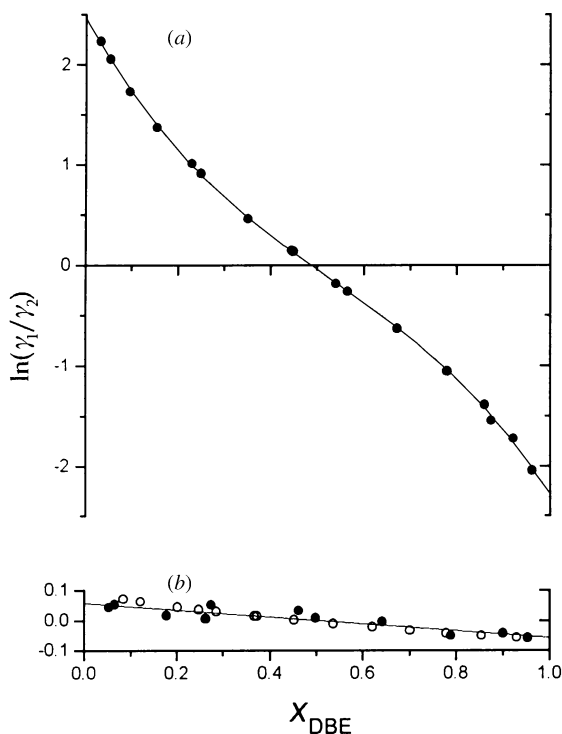


Fig. 1 Plot of $\ln(\gamma_1/\gamma_2)$ vs. x_{DBE} for: (a) DBE(1) + methanol(2); (b) DBE(1) + hexane(2) at 298.15 K. (●) experimental; (—) calculated from eqn. (4) with coefficients from Table 2; (○) Marsh *et al.* (ref. 12).

By examining Table 3 it can be noticed that binary mixtures of DBE with hexane, octane and diethyl ether behave almost ideally. The other systems exhibit positive deviations, *i.e.* $G^E > 0$ and $\gamma_i > 1$, which roughly increase with increasing polarity of component 2. The lowest deviations are displayed by mixtures with hydrocarbons and ethers, the largest by mixtures with alcohols and acetonitrile. The G^E and γ_i values for DBE + an alcohol and DBE + an amine seem to decrease for increasing size of the hydrocarbon portion of component 2.

The γ_i^∞ values of organic compounds in octane, and in general in hydrocarbons,²⁷ show a dependence on the solute polarity which is similar, though more pronounced, to that observed in DBE. This is evidence of the weakly polar character of the DBE solvent.

The effect of breaking and/or forming hydrogen bonds, and in general of association phenomena in solution, is emphasized by the activity coefficients of the components, γ_i , which are a measure of their tendency to escape from the solution. For all mixtures examined here the γ_i of both components

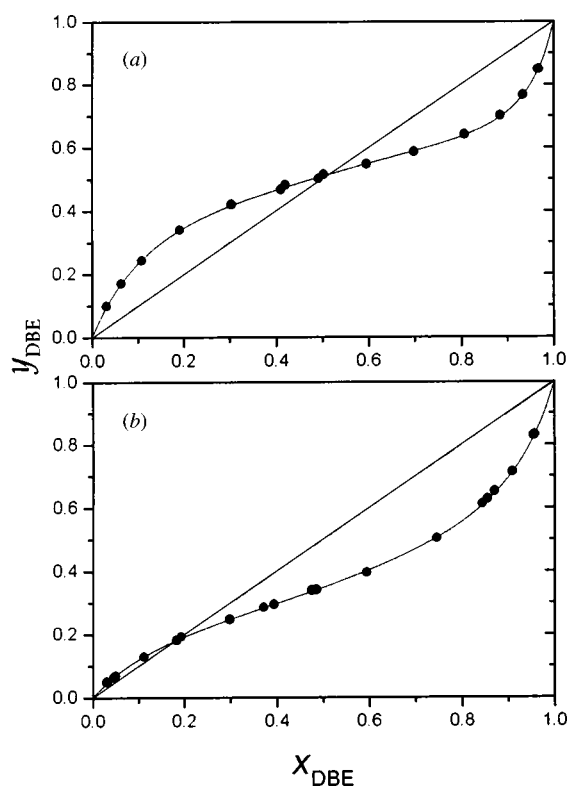


Fig. 2 Plot of y_{DBE} vs. x_{DBE} for: (a) DBE + butanol; (b) DBE + cyclopentanone at 298.15 K. (●) experimental; (—) calculated from eqn. (10) with parameters from Table 2.

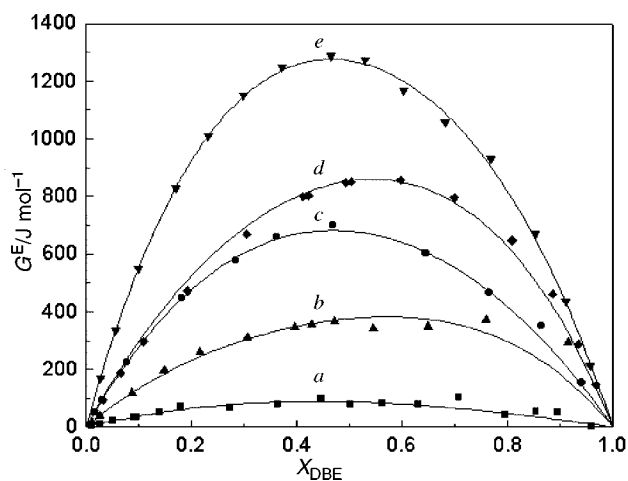


Fig. 3 Excess Gibbs energies G^E vs. x_{DBE} for DBE + an organic compound at 298.15 K. Compounds: (a) tetrahydrofuran; (b) propylamine; (c) propanone; (d) butan-1-ol; (e) acetone. (—) calculated from eqn. (6); experimental points from eqn. (8).

increase with dilution reaching a maximum at infinite dilution. The increase is steeper the more polar the second component. This behaviour is evidence for self-association, which is greatest in the pure liquid (low escaping tendency) and lowest in infinitely dilute solution (high escaping tendency).

According to the simple interpretative scheme proposed by Patterson,²⁸ the positive G^E , H^E and TS^E values for multi-component liquid mixtures can be qualitatively explained with a net destruction, in the mixing process, of the structural order present in the pure liquids, *i.e.*, H-bonding in alcohols and amines, dipolar interactions for other polar compounds. This interpretation applies to most of the examined systems containing DBE. However, for methanol and butanol in DBE, and in general for alcohols in inert solvents, positive H^E values²⁹ reveal a net breaking of H-bonds, whereas negative TS^E and V^E ²⁰ indicate a more ordered structure in solution. These apparently conflicting results are hard to explain. The basic hypothesis that order destruction/formation brings about positive/negative Y^E ($Y = H, S, G, V$) values is evidently an oversimplification, order not being a well defined property. Thus, the Patterson scheme is inadequate in some cases, though being a useful tool for a qualitative interpretation of the sign of the excess quantities for most mixtures. Insights into the structure of the investigated solutions may come from calculation of the fluctuation of Kirkwood–Buff integrals, G_{ij} .¹⁷

Application of the fluctuation theory of mixtures

The Kirkwood–Buff integrals, G_{ij} , provide a measure of the affinity or tendency to association of species i and j in solu-

Table 3 Excess Gibbs energies for equimolar mixtures, G^E ($x = 0.5$), and activity coefficients at infinite dilution, γ_i^∞ , for the components of binary mixtures dibutyl ether (1) + an organic compound (2) at 298.15 K

Compound (2)	G^E ($x = 0.5$)/ J mol^{-1}	γ_1^∞	γ_2^∞
Hexane	35 ± 7	1.06	1.06
	50.4 ^a	1.106 ^a	1.068 ^a
Octane	42 ± 3	1.07	1.07
Diethyl ether	5 ± 4	1.01	1.01
	2 ^b	1.004 ^b	1.003 ^b
Tetrahydrofuran	87 ± 5	1.19	1.11
Propylamine	376 ± 13	1.88	2.47
Butylamine	318 ± 10	1.66	2.35
Propanone	680 ± 6	3.70	2.89
Cyclopentanone	596 ± 6	3.05	2.82
Methanol	1241 ± 8	11.8	9.77
	—	10.56 ^c	—
Butan-1-ol	854 ± 8	3.9	6.02
	—	2.54 ^d	—
	888 ^e	3.9 ^e	3.5 ^e
	318 ^f	2.7 ^f	1.3 ^f
	603 ^g	2.5 ^g	2.1 ^g
Acetonitrile	1271 ± 6	14.6	9.3

^a Marsh *et al.* (ref. 12), isothermal data at 298.15 K. ^b Papoušek and Smékal (ref. 8), isobaric data at 600 mmHg. Computed from the reported van Laar parameters. ^c Delcros *et al.* (ref. 13), isothermal data at 333.15 K. ^d Delcros *et al.* (ref. 13), isothermal data at 373.15 K. ^e Greune and Kehlen (ref. 9), isobaric data at 50 mmHg. Evaluated from Margules constants reported in ref. 25. ^f Smirnova and Morachevsky (ref. 10), isobaric data at 760 mmHg. Evaluated from van Laar constants reported in ref. 25. ^g Volkova *et al.* (ref. 11), isobaric data at 760 mmHg. Evaluated from Margules constants reported in ref. 25.

tion,^{17,30} and can be used, among others, to estimate local composition and preferential solvation.^{31,32} The equations and the procedure for the calculation of the G_{ij} have been reported in previous papers.^{17,30–32} The thermodynamic quantities required in the calculations were taken from this work (G^E) and from other results from our laboratory (V^E),²⁰ the isothermal compressibilities of the mixtures, k_T , were calculated from those of the neat components assuming a zero excess k_T value.³³

The results that we have deemed most relevant to the present discussion are reported in Table 4 together with other quantities useful for these calculations. They are the maximum value of the affinity between species 2, $\Delta G_{22}^{\text{max}}$, its value at infinite dilution, ΔG_{22}^∞ , and $\Delta n_{22}^{\text{max}}$, that is the maximum value of the excess of molecules of species 2 in the neighbourhood of a central molecule of the same species. The Δ before these symbols indicate that, as discussed in previous papers,³⁴ the quantities have been calculated with respect to the ideal refer-

Table 4 Kirkwood–Buff integrals and related quantities at 298.15 K

Component 2	h^a	$\Delta G_{22}^{\text{max}}/$ $\text{cm}^3 \text{mol}^{-1}$	$\Delta G_{22}^\infty/$ $\text{cm}^3 \text{mol}^{-1}$	$\Delta n_{22}^{\text{max}}$	$b^b/$ kJ mol^{-1}	$g_T^c/$ $\text{cm}^3 \text{mol}^{-1} \text{K}^{-1}$
Hexane	−0.11	20	20	0.02	−2.52	−1.1
Octane	−0.14	22	22	0.02	—	—
Diethyl ether	−0.02	4	4	0.04	−0.56	−0.3
Tetrahydrofuran	−0.07	20	11	0.07	−2.31	−1.1
Propylamine	−3.65	620	620	0.5	−2.80	−0.6
Butylamine	−3.72	633	630	0.5	−2.20	−0.3
Propanone	−2.33	390	390	1.0	−15.47	−6.7
Cyclopentanone	−2.83	483	480	0.6	−16.77	−7.2
Methanol	−7.13	2120	1220	10	−44.43	−19
Butan-1-ol	−6.06	1360	1040	2.0	−49.16	−21
Acetonitrile	−6.77	5090	1200	43	−30.14	−13

^a $h = (\ln \gamma_2/dx_2)_{x_2=0}$. ^b $b = (d\Delta_{\text{soln}}H/dx_2)_{x_2=0}$. ^c $g_T = d\Delta G_{22}^\infty/dT$.

ence state. We have not reported any figure or table with information on the ΔG_{22} and Δn_{22} behaviour in the whole mole fraction range because for all systems ΔG_{22} shows the same sign as $\Delta G_{22}^{\max}$ and $\Delta n_{22}^{\max}$ values are all in the range $0.4 < x_2 < 0.6$. Also, due to relations among the ΔG_{ij} quantities,^{33,34} ΔG_{11} has the same sign as ΔG_{22} and ΔG_{12} the opposite one.

Table 4 shows that $\Delta G_{22}^{\max}$ values are all positive, indicating that the like-like interactions and tendency to association are the dominant ones, though with different strength. Mixtures with hydrocarbons and ethers have the lowest $\Delta G_{22}^{\max}$, their values not exceeding $50 \text{ cm}^3 \text{ mol}^{-1}$, and $\Delta n_{22}^{\max}$ less than 0.1. Ketones and amines show a moderate strength of like-like interactions, as witnessed by $\Delta G_{22}^{\max}$ values of the order of some hundreds and $\Delta n_{22}^{\max}$ ranging from 0.5 to 1. Alcohols and acetonitrile show a very strong like-like affinity, with $\Delta G_{22}^{\max}$ values larger than a thousand; in particular, the latter system and to a lesser extent the system DBE + methanol, are near a critical solution temperature, as the very large $\Delta n_{22}^{\max}$ values suggest.

The ΔG_{22}° values, which in some cases coincide with $\Delta G_{22}^{\max}$, indicate that the like-like affinity is effective also in the limit of infinite dilution of component 2. For these quantities, an estimation of the temperature dependence coefficient $g_T = d\Delta G_{22}^\circ/dT$ can be obtained by differentiating the expression for ΔG_{22}° .³⁴

$$\Delta G_{22}^\circ = 2(V_2^* - \bar{V}_2^\circ) - V_1^* \left(\frac{d \ln \gamma_2}{dx_2} \right)_{x_2=0} \simeq -V_1^* h \quad (11)$$

where h is the limiting slope of the $\ln \gamma_2$ curve. In the second part of the equation the approximation has been introduced that the former term on the right hand side is negligible with respect to the latter one, as happens for all our systems except maybe DBE + diethyl ether. We obtain then:

$$g_T = \frac{V_1^*}{RT^2} \left(\frac{d\bar{H}_2}{dx_2} \right)_{x_2=0} - V_1^* \alpha_1 \left(\frac{\ln \gamma_2}{dx_2} \right)_{x_2=0} = \frac{2V_1^*}{RT^2} b - V_1^* \alpha_2 h \quad (12)$$

where α_1 is the coefficient of thermal expansion of DBE and b is the limiting derivative with respect to x_2 of the enthalpy of solution, $\Delta_{\text{soln}} H$, of component 2 in DBE, whose values are reported elsewhere.³⁵ The results in Table 4 show that g_T values are all negative, suggesting that the like-like affinity decreases with increasing temperature. Moreover, their magnitude is larger the larger the ΔG_{22}° magnitude, so that for most systems a change of sign of ΔG_{22}° can be expected for a temperature increase of about 50 K.

The clear-cut correlation shown by ΔG_{22}° and b values (see Table 4) indicates that the like-like attraction at very high dilution is the result of negative enthalpies of formation of the 2–2 couple, which are the largest in magnitude for the alcohols and very large also for polar compounds such as ketones and acetonitrile. Amines show an unexpected very low magnitude of b , similar to that for hexane and tetrahydrofuran, practically suggesting the absence of H-bonding in the 2–2 interaction in very dilute solution.

Conclusions

The excess Gibbs energies and activity coefficients of components reveal positive deviations for all mixtures here investigated, the non-ideality increasing with the polarity of the second component.

The model of Patterson, based on the concept of order formation/destruction, can explain the positive sign of G^E , H^E , S^E , V^E observed for most systems, while fails for

DBE + methanol and DBE + butanol which show negative S^E , V^E and positive G^E , H^E .

The application of fluctuation theory indicates that the tendency to association of solutes in DBE, as measured by the Kirkwood–Buff integrals ΔG_{22} , increase in the order: hydrocarbons, ethers, ketones, amines, alcohols, acetonitrile.

At high dilution, the like-like affinity decreases with increasing temperature. A correlation has been found between ΔG_{22}° and the limiting slope of the enthalpy of solution, indicating that the like-like attraction is the result of a negative enthalpy of formation of the 2–2 couple.

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List of symbols

Latin

a	Redlich–Kister parameter
A	chromatographic peak area
b	limiting slope of the solution enthalpy
B_{ij}	virial coefficient
g_T	temperature coefficient of Kirkwood–Buff integral
G^E	molar excess Gibbs Energy
G_{ij}	Kirkwood–Buff integral
h	limiting slope of activity coefficient
$\bar{H}$	partial molar enthalpy
H^E	molar excess enthalpy
$\Delta_{\text{soln}} H$	enthalpy of solution
p^*	vapour pressure of pure liquid
P	total vapour pressure of mixture
r_f	chromatographic response factor
R	gas constant
S	objective function
S^E	molar excess entropy
T	absolute temperature
$\bar{V}$	partial molar volume
V^*	molar volume of pure substance
V^E	molar excess volume
W	correction term for vapour non-ideality
x	liquid molar fraction
y	vapour molar fraction

Greek

α	coefficient of thermal expansion
γ	activity coefficient
κ_T	isothermal compressibility
σ	standard deviation

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